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Does Dimensionality curse effect some models more than others?

Asked 6 years, 10 months ago Modified 3 years, 3 months ago Viewed 13k times



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The places I have been reading about dimensionality curse explain it in conjunction to kNN primarily, and linear models in general. I regularly see top rankers in Kaggle using thousands of features on dataset that hardly has 100k data points. They mainly use Boosted trees and NN, among others. That many features seem too high and I feel they would get affected by dimensionality curse. But that doesn't seem to be the case as these models make them top the competitions. So, back to my original question - do some models get affected by dimensionality curse more than others?

Specifically, I am interested in the following models (only because these are the ones that I am aware of/used):

- Linear and Logistic Regression
- Decision Trees/RandomForest/Boosted Trees
- Neural Networks
- SVM
- kNN
- k-means clustering

neural-networks

svm

k-means

k-nearest-neighbour

high-dimensional

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edited Jul 15, 2019 at 5:53



ttnphns

53.5k

42

262

480

asked Dec 11, 2015 at 1:29



Dileep Kumar

Patchigolla

781

2

9

17



The short answer is definitely yes, but maybe you want models you are actually interested in? I'm sure the CV community could tell you about thousands of different types of models that are affected by the curse of dimensionality. So narrowing your focus to certain type of models may help with answering this question. – user95564 Dec 11, 2015 at 3:45



@RustyStatistician - I have added a few models I am interested in. – Dileep Kumar Patchigolla Dec 11, 2015 at 4:30



I am quite interested in this question but it remained unanswered. How can I bring this up in visibility, to get answers? – Dileep Kumar Patchigolla Dec 15, 2015 at 11:07

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1 Answer

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Generalized Linear Models



Linear models suffer immensely from the curse of dimensionality. Linear models partition the space in to a single linear plane. Even if we are not looking to directly compute



$$\hat{\beta} = (X'X)^{-1}X'y$$

the problem posed is still very sensitive to collinearity, and can be considered "ill conditioned" without some type of regularization. In very high dimensional spaces, there is more than one plane that can be fitted to your data, and without proper type of regularization can cause the model to behave very poorly. Specifically what regularization does is try to force one unique solution to exist. Both L1 and squared L2 regularization try to minimize the weights, and can be interpreted selecting the model with the smallest weights to be the most "correct" model. This can be thought of as a mathematical formulation of Occams Razor.

Decision Trees

Decision trees also suffer from the curse of dimensionality. Decision trees directly partition the sample space at each node. As the sample space increases, the distances between data points increases, which makes it much harder to find a "good" split.

Random Forests

Random Forests use a collection of decision trees to make their predictions. But instead of using all the features of your problem, individual trees only use a subset of the features. This minimizes the space that each tree is optimizing over and can help combat the problem of the curse of dimensionality.

Boosted Tree's

Boosting algorithms such as AdaBoost suffer from the curse of dimensionality and tend to overfit if regularization is not utilized. I won't go in depth, because the post [Is AdaBoost less or more prone to overfitting?](#) explains the reason why better than I could.

Neural Networks

Neural networks are weird in the sense that they both are and are not impacted by the curse of dimensionality dependent on the architecture, activations, depth etc. So to reiterate the curse of dimensionality is the problem that a huge amount of points are necessary in high dimensions to cover an input space. One way to interpret deep neural networks is to think of all layers except the very last layer as doing a complicated projection of a high dimensional manifold into a lower dimensional manifold, where then the last layer classifies on top of. So

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sense the compressed representation of our data allows us to circumvent the curse of

dimensionality. Again this is one interpretation, in reality the curse of dimensionality does in fact impact neural networks, but not at the same level as the models outlined above.

SVM

SVM tend to not overfit as much as generalized linear models due to the excessive regularization that occurs. Check out this post [SVM, Overfitting, curse of dimensionality](#) for more detail.

K-NN, K-Means

Both K-mean and K-NN are greatly impacted by the curse of dimensionality, since both of them use the L2 squared distance measure. As the amount of dimensions increases the distance between various data-points increases as well. This is why you need a greater amount of points to cover more space in hopes the distance will be more descriptive.

Feel free to ask specifics about the models, since my answers are pretty general. Hope this helps.

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edited Apr 13, 2017 at 12:44



Community Bot

1

answered Dec 15, 2015 at 19:10



Armen Aghajanyan

1,023 8 12

▲ Hi Amen Great succinct explanations for all the models I have asked. Issues with linear models is still not clear for me: Do linear models perform better or worse than k-NN and k-Means models for the same no:of dimensions? And when you said collinearity is an issue for linear models, do you imply that with no (or minimal) collinearity, high dimensions is not an issue with linear models?
– [Dileep Kumar Patchigolla](#) Dec 16, 2015 at 5:54

▲ It is hard to quantify if linear models will perform better than k-nn or k-means for an arbitrary problem. If your problem is linearly separable, I would place my bets on the linear model, while if your space is a bit more complicated, I would go with k-nn. Collinearity worsens the problem of the curse of dimensionality, even without collinearity, the curse of dimensionality still applies. K-means should suffer to the same extent as k-nn as they both are neighbor driven, and generally use the same distance function. In reality it is hard to quantify how bad the COD is. Hope this helps!
– [Armen Aghajanyan](#) Dec 16, 2015 at 6:20

▲ What is your definition of curse of dimensionality (CoD)? Your answer seems to suggest that linear models suffer the most from CoD, this is misleading: being a global method, linear models suffer much less than localized methods such as KNN. – [Matifou](#) Feb 24, 2019 at 19:08